CAT Quant formula sheet

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Algebra

- $(a + b)^2 = a^2 + 2ab + b^2$
- $a^2 - b^2 = (a - b)(a + b)$
- $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
- $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$ If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$.
- $ax^2 + bx + c = 0$: $x = [-b \pm \sqrt{(b^2 - 4ac)}] / 2a$ Sum of roots = $-b/a$, product = c/a.
- AM ≥ GM: $(a + b)/2 \geq \sqrt{(ab)}$ for $a, b \geq 0$ Equality when $a = b$.

Progressions

- AP: $T_n = a + (n - 1)d$, $S_n = n/2\ [2a + (n - 1)d]$
- GP: $T_n = ar^{n-1}$, $S_n = a(r^n - 1)/(r - 1)$ $r \neq 1$.
- Infinite GP: $S = a/(1 - r)$ $|r| < 1$.
- $1 + 2 + \ldots + n = n(n + 1)/2$
- $1^2 + 2^2 + \ldots + n^2 = n(n + 1)(2n + 1)/6$
- $1^3 + 2^3 + \ldots + n^3 = [n(n + 1)/2]^2$

Logarithms

- $\log(mn) = \log m + \log n$, $\log(m/n) = \log m - \log n$
- $\log m^k = k \log m$
- $\log_a b = \log b / \log a$ Change of base.
- $a^{(\log_a x)} = x$

Number system

- $N = p^a q^b r^c$ has $(a + 1)(b + 1)(c + 1)$ factors
- Sum of factors = $[(p^{a+1} - 1)/(p - 1)] \times \ldots$ Multiply over every prime.
- HCF × LCM = product (two numbers)
- Trailing zeros of $n!$ = $\lfloor n/5 \rfloor + \lfloor n/25 \rfloor + \lfloor n/125 \rfloor + \ldots$
- Euler: $a^{\varphi(n)} \equiv 1 \pmod n$ if $\gcd(a, n) = 1$ $\varphi(p) = p - 1$ for a prime p.

Counting & probability

- ${}^nP_r = n!/(n - r)!$, ${}^nC_r = n!/[r!(n - r)!]$
- Circular arrangements of $n = (n - 1)!$
- Ways to split n identical items among r people = ${}^{n+r-1}C_{r-1}$ Each may get zero.
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(\text{not } A) = 1 - P(A)$

Geometry & mensuration

- Triangle: area = $\frac{1}{2}ab \sin C = \sqrt{[s(s - a)(s - b)(s - c)]}$ $s = (a + b + c)/2$ (Heron).
- Equilateral triangle area = $(\sqrt{3}/4)a^2$
- Interior angle of a regular n-gon = $(n - 2) \times 180°/n$
- Diagonals of an n-gon = $n(n - 3)/2$
- Circle: area πr^2, circumference $2\pi r$; sector = $(\theta/360)\pi r^2$
- Sphere $V = (4/3)\pi r^3$, $S = 4\pi r^2$; cone $V = (1/3)\pi r^2 h$; cylinder $V = \pi r^2 h$